

Task 3: Basic SELECT Queries

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Role: SQL Developer Intern

Task Overview & Metadata

Objective: Extract and manipulate data from database tables using standard SQL clauses.

Tools Used: DB Browser for SQLite / MySQL Workbench

Key Concepts: **SELECT** **WHERE** **ORDER BY** **LIMIT** **LIKE** **BETWEEN**
DISTINCT

1. Conceptual Assessment & Answers

1. What does SELECT do?

The SELECT statement is used to retrieve data records from one or more database tables and project specified columns in the result set.

2. How do you filter rows?

Rows are filtered using the WHERE clause followed by condition statements involving operators such as =, >, <, LIKE, IN, and BETWEEN.

3. What is LIKE '%value%'?

The LIKE operator with the %value% wildcard pattern searches for text containing the substring "value" anywhere within the target column, regardless of preceding or trailing characters.

4. What is BETWEEN used for?

The BETWEEN operator filters data within an inclusive range, matching values greater than or equal to the low boundary and less than or equal to the high boundary.

5. How do you limit output rows?

By appending the LIMIT clause at the end of a query followed by a numerical value (e.g., LIMIT 10 restricts the output to 10 rows).

6. Difference between = and IN

The = operator checks for exact equality against a **single scalar value**, whereas the IN operator evaluates whether a value matches **any element in a list** of multiple values.

7. How to sort in descending order?

Use the ORDER BY clause followed by the column name and the DESC keyword (e.g., ORDER BY salary DESC).

8. What is aliasing?

Aliasing assigns a temporary name to a table or column in the query results using the AS keyword, improving readability and clarifying expression output.

9. Explain DISTINCT.

The DISTINCT keyword removes duplicate rows from the query result set, returning only unique combinations of values.

10. What is the default sort order?

The default sort order in an ORDER BY clause is **Ascending (ASC)**.

2. SQL Script Deliverable

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-- =====
-- Elevate Labs: SQL Developer Internship
-- Task 3: Writing Basic SELECT Queries
-- Student Name: G. Pavana Deepika
-- =====

-- 1. Use SELECT * and specific columns
SELECT * FROM employees;

SELECT employee_id, first_name, last_name, salary, department_id
FROM employees;

-- 2. Apply WHERE, AND, OR, LIKE, BETWEEN filtering
-- Filter with WHERE & AND
SELECT * FROM employees
WHERE department_id = 3 AND salary > 50000;

-- Filter with OR
SELECT * FROM employees
WHERE department_id = 1 OR department_id = 2;

-- Pattern matching with LIKE
SELECT * FROM employees
WHERE first_name LIKE '%an%';

-- Range filtering with BETWEEN
```

```
SELECT * FROM employees
WHERE salary BETWEEN 40000 AND 80000;

-- 3. Sort with ORDER BY and limit output with LIMIT
SELECT employee_id, first_name, salary
FROM employees
WHERE salary > 30000
ORDER BY salary DESC
LIMIT 5;

-- 4. Column Aliasing and Unique Values using DISTINCT
SELECT DISTINCT department_id AS unique_department_id
FROM employees;
```